SOFT COMPUTING · MINI PROJECT

SIGNAL LAB

Retiming a city’s traffic lights with GA, ACO, PSO and SA, plus two methods from beyond the syllabus



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WHAT WE DID

From a pretend city to a live race, in four steps

1

Built a city

A traffic simulator: 9 crossroads, 15 minutes of rush hour. It scores any set of light timings by the average wait per car.

2

Coded 6 brains

GA, ACO, PSO and SA from our syllabus, plus CMA-ES and Max-Pressure, all written from scratch, all tuning the same 18 dials.

3

Ran a fair race

3,000 tries each, repeated 10 times with different random starts, on 3 different traffic patterns.

4

Made it visible

A live web app: watch every method learn, open any one up, change its settings and race again.

50.7 s → 28.2 s Our best plans cut the average wait at red lights by 44% compared with ordinary, untuned lights.

Sudiksha Singh · R066

THE PROBLEM

A signal plan is a hard search problem

18

variables

Nine junctions, each with a green split and an offset. Every combination is a different city.

0

gradients

Delay has no formula. The only way to score a plan is to simulate the traffic.

∞

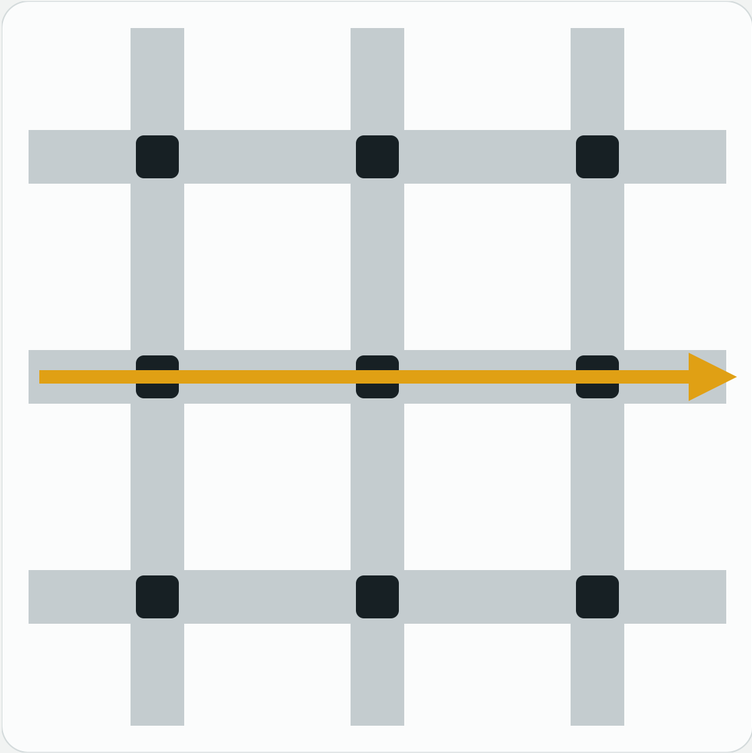
local optima

Offsets interact. A green wave only works when neighbouring junctions line up.

Sudiksha Singh · R066

THE SIMULATED CITY

Nine junctions, 15 minutes of peak traffic



Element	Setting
Signals	60 s cycle, 2 phases, 3 s amber + all-red
Capacity	1800 veh/h per lane on green
Links	8 s travel time between junctions
Turning	76% straight, 12% left, 12% right
Demand	Peak profile from the grid edges
Traffic flow	Store-and-forward queues, left-hand driving

Main scenario: an arterial rush hour, with heavy eastbound commuters on the middle road (the amber arrow).

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ENCODING AND FITNESS

Every algorithm sees the same 18 numbers

J1	J2	J3	J4	J5	J6	J7	J8	J9
split	split	split	split	split	split	split	split	split
offset	offset	offset	offset	offset	offset	offset	offset	offset

Split ∈ [0, 1] maps to 9–45 s of north–south green. Offset ∈ [0, 1) maps to 0–60 s and wraps around the cycle.

Fitness (minimise)

Average delay = total vehicle-seconds spent queuing ÷ vehicles entering. Vehicles still stuck at the end cost 90 s each.

A fair race

One evaluation = one 900 s simulation (≈ 0.4 ms). Every algorithm gets the same budget: 3,000 evaluations.

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LIVE DEMO

Watch the algorithms race in real time

Six live feeds

Each tile simulates the best plan its algorithm has found so far. Queues shrink as the search improves.

Convergence race

Best delay against evaluations, with a leaderboard against the untuned plan and Max-Pressure.

Expand any algorithm

See its search space, its timing plan and its internal state, then change its parameters and rerun it.

Open: Signal Lab app

ALGORITHM 1 OF 4 · IN SYLLABUS

Genetic Algorithm

A chromosome is a whole timing plan. Fitter plans breed, so good green waves found in different plans get combined.

Parameter	Value
Population	30
Crossover	BLX-α, rate 0.9
Mutation	0.08 per gene, σ 0.15
Selection	Tournament of 3, 2 elites

One generation

1. Simulate every plan in the population
2. Copy the 2 best plans unchanged
3. Pick parents by tournament
4. Blend parent genes (BLX-α)
5. Mutate with Gaussian noise; wrap offsets

Mean delay, 10 seeds

28.33 s

Spread (SD)

± 0.17 s

Evals to 95% of its gain

312

ALGORITHM 2 OF 4 · IN SYLLABUS

Ant Colony Optimisation (MAX-MIN)

Each split and offset is cut into 12 levels. An ant’s path through 18 choice points is a timing plan.

Parameter	Value
Ants per iteration	15
Levels per gene	12 (5 s steps)
Evaporation ρ	0.08
Pheromone bounds	[0.05, 4], α = 1

One iteration

1. Each ant picks a level per gene, p ∝ τ^α
2. Simulate every ant’s plan
3. Evaporate all pheromone by ρ
4. Reinforce iteration-best and best-so-far
5. Clamp τ between the MAX-MIN bounds

Mean delay, 10 seeds

29.59 s

Spread (SD)

± 0.54 s

Evals to 95% of its gain

711

ALGORITHM 3 OF 4 · IN SYLLABUS

Particle Swarm Optimisation

Each particle is a timing plan flying through 18-D space, pulled by its own best and the swarm's best.

Parameter	Value
Swarm size	24
Inertia w	0.72
$c_1 = c_2$	1.49
Max velocity	0.2 per step

One step per particle

$$v \leftarrow w \cdot v + c_1 r_1 (pbest - x) + c_2 r_2 (gbest - x)$$

1. Update velocity, clamp to ±vmax
2. Move; wrap offsets into one cycle
3. Simulate the new plan
4. Update personal and swarm bests

Mean delay, 10 seeds

28.45 s

Spread (SD)

± 0.69 s

Evals to 95% of its gain

296 · fastest

ALGORITHM 4 OF 4 · IN SYLLABUS

Simulated Annealing

One plan, nudged a few genes at a time. Worse moves are sometimes accepted so the search can escape local minima.

Parameter	Value
Temperature	3 → 0.01, geometric
Step size	0.18 → 0.02
Genes per move	1 to 3
Population	1 (single point)

One move

1. Perturb 1–3 genes with Gaussian noise
2. Simulate the neighbour
3. Better: accept. Worse: accept with p = e^(−Δ/T)
4. Cool T and shrink the step

Mean delay, 10 seeds

28.21 s · best mean

Spread (SD)

± 0.11 s

Evals to 95% of its gain

1313 · slowest

BEYOND THE SYLLABUS

Two methods our course doesn’t teach

CMA-ES

An evolution strategy that learns how the 18 variables interact.

Max-Pressure

A real-time controller that needs no optimisation at all.

Kevin Pratap · R064

BEYOND SYLLABUS 1 · HANSEN & OSTERMEIER, 2001

CMA-ES: Covariance Matrix Adaptation

Samples plans from a Gaussian, then reshapes that Gaussian towards the best ones.

- Learns the full 18×18 covariance, so it finds coupled offsets that form green waves
- Adapts its own step size σ from the evolution path
- Only two settings: λ = 12 samples, σ_0 = 0.25

One generation

1. Sample λ plans: x = m + σ·B·D·z
2. Simulate and rank them
3. Move mean m to the weighted best μ = 6
4. Update covariance (rank-one + rank-μ)
5. Grow or shrink σ by path length

Best single run

28.03 s

Mean ± SD, 10 seeds

28.95 ± 1.32 s

Balanced grid

Best of all, 28.71 s

BEYOND SYLLABUS 2 · VARAIYA, 2013

Max-Pressure: decide every second

No plan to optimise. Each junction reads its live queues and serves the phase with the most pressure.

pressure(phase) = Σ (queue upstream − queue it feeds downstream)

Switch when the other phase leads by more than 1 vehicle and 10 s of minimum green have passed.

Provably throughput-optimal, but local: it never plans a green wave.

Scenario	Max-Pressure	Best optimiser
Balanced grid	20.98 s, wins	28.71 s
Arterial rush	31.24 s	28.21 s, wins
Morning inbound	36.76 s	29.80 s, wins

On the balanced grid, Max-Pressure’s delay was **27% lower** than the best tuned fixed plan.

RESULTS · ARTERIAL RUSH HOUR · 10 SEEDS × 3,000 EVALUATIONS

Every optimiser cut delay by over 40%

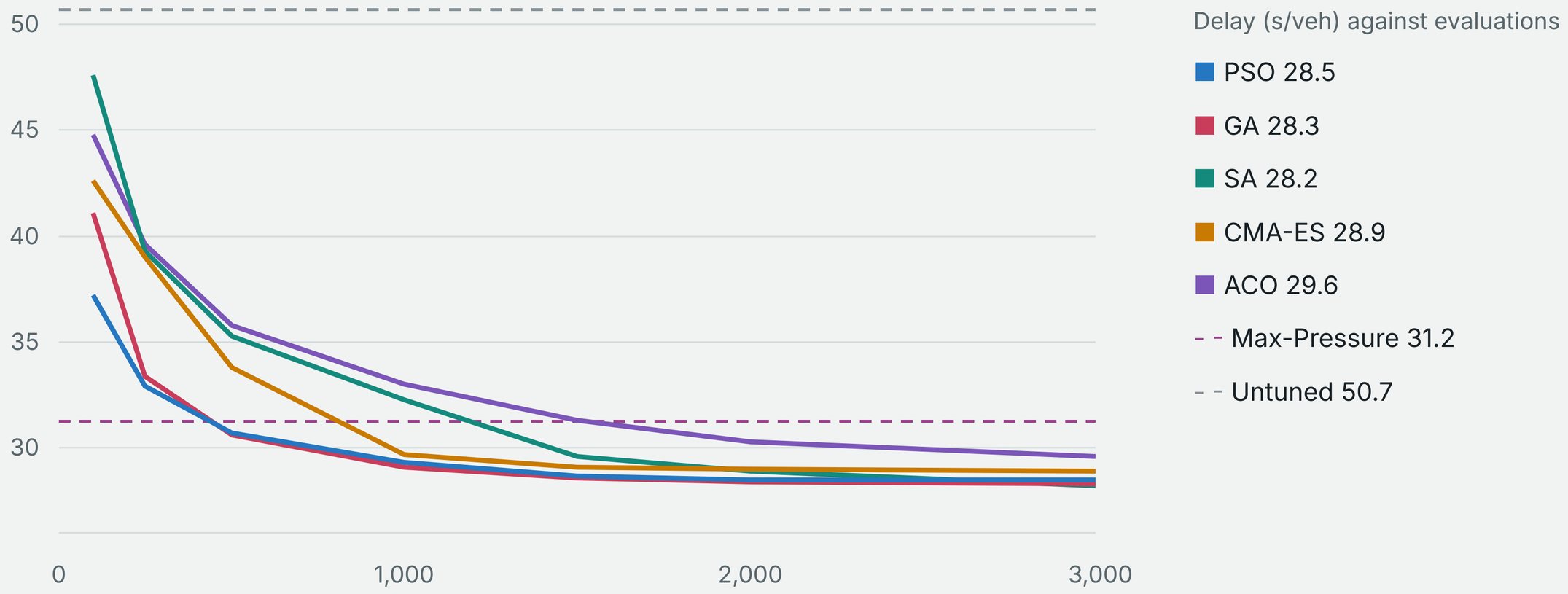
Controller	Mean	SD	Best	Worst	vs 50/50	Evals 95%
Simulated Annealing	28.21	0.11	28.06	28.40	−44.4%	1313
Genetic Algorithm	28.33	0.17	28.15	28.69	−44.1%	312
Particle Swarm	28.45	0.69	28.02	30.22	−43.9%	296
CMA-ES	28.95	1.32	28.03	31.58	−42.9%	525
Ant Colony	29.59	0.54	28.84	30.49	−41.6%	711
Max-Pressure	31.24	–	–	–	−38.4%	live
Fixed 50/50, untuned	50.70	–	–	–	–	–

Average delay in seconds per vehicle. Evals 95% = median evaluations to reach 95% of the run’s final improvement.

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CONVERGENCE · MEAN OF 10 SEEDS · ARTERIAL

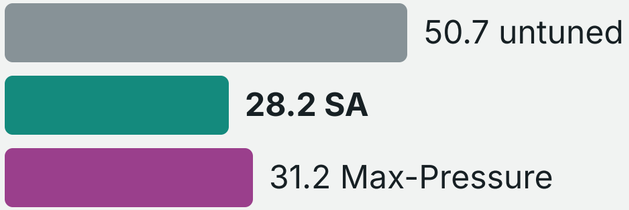
Fast starters versus slow finishers



THREE TRAFFIC PATTERNS · AVERAGE DELAY, S PER VEHICLE

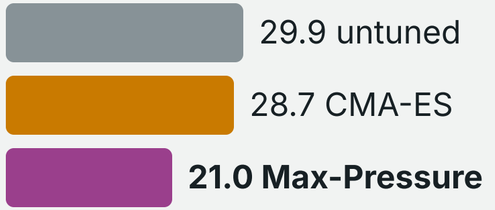
The winner depends on the traffic

Arterial rush



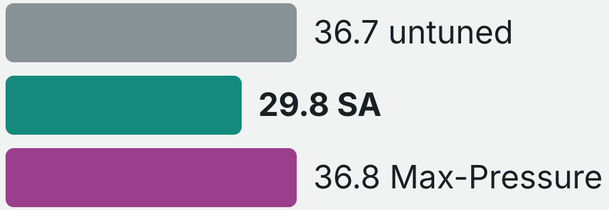
Green waves on one dominant corridor favour a coordinated fixed plan.

Balanced grid



No corridor to coordinate, so reacting to live queues wins by 27%.

Morning inbound



One saturated corner. Max-Pressure loses green time switching; the tuned plan doesn’t.

Bars to one scale. Best optimiser shown per scenario, mean of 10 seeds.

CONCLUSIONS

What we learned

Tuning beats algorithm choice

All five cut delay by 42–44%. The top four end within 0.8 s of each other.

Speed and quality trade off

PSO and GA converge in ~300 evaluations. SA finishes best but needs 4× longer.

Representation matters

ACO came last because discretising timings into 5 s steps limits fine-tuning.

No single winner

Adaptive control wins on balanced demand; coordinated plans win on corridors.

Next: a hybrid that optimises offsets offline and adapts splits live, IPOP-CMA-ES restarts, and testing on a real network in SUMO.

THANK YOU

QUESTIONS?

Try it yourself: Signal Lab app

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Intro, problem, the city

Kevin Pratap · R064
Demo, GA, ACO, PSO, SA

Shlok Shah · R059
CMA-ES, Max-Pressure, results